

# Allison Liu

azn@allisonznliu.com

LinkedIn: allisonznliu

Portfolio: allisonznliu.com

Data Scientist with federal transportation experience translating quantitative analysis and geospatial data into decision-ready tools. Skilled in Python-based GIS workflows, data visualization, and data management.

## SKILLS

**GIS & Geospatial:** QGIS, ArcGIS, GDAL, Git, Geospatial Data Processing, Spatial Analysis

**Programming & Analysis:** Python, Git, SQL, Data Visualization, Machine Learning (PyTorch, TensorFlow, scikit-learn), Image Analysis

**Web & Tools:** HTML/CSS/JavaScript, MS Excel/Word/PowerPoint

## EXPERIENCE

### Data Scientist

**April 2024 – September 2025**

*Bureau of Transportation Statistics, U.S. Department of Transportation*

*Washington, DC*

- Built a nationally consistent geospatial dataset identifying transportation infrastructure exposed to natural hazards. Developed a custom web map enabling stakeholders to prioritize critical assets for analysis.
- Led collaboration with 50+ stakeholders (including state DOTs, local governments, researchers, and industry) to develop a national data standard for bike, pedestrian, and accessibility infrastructure.
- Automated geospatial data processing pipelines for federal transportation noise datasets, enabling migration to a lower-cost hosting platform, improving user interface, and increasing the accuracy, reproducibility, and efficiency of statistical updates.

### Applied Mathematics Graduate Research Assistant

**May 2022 – May 2023**

*University of Colorado Boulder*

*Boulder, CO*

- Generated novel features to improve solar flare prediction models by applying machine learning, image analysis, and dimension reduction techniques on terabyte-scale solar image data. Extracted meaningful spatial/temporal trends from large, noisy datasets.

### Student Data Scientist

**February 2021 – December 2022**

*Laboratory for Atmospheric and Space Physics (LASP)*

*Boulder, CO*

- Created a machine learning ready dataset for solar flare prediction researchers by integrating 17 years of NASA magnetogram datasets using generative adversarial networks (GANs).

### Student Research Assistant

**June 2017 – August 2020**

*Kapteyn-Murnane Group, JILA, University of Colorado Boulder*

*Boulder, CO*

- Developed computational image-reconstruction algorithms for laser beam profile diagnostics. Work contributed to a publication in Science Magazine.

### Climbing Gym Route Setter at University of Colorado Boulder

**September 2020 – August 2022**

- Collaborated closely with a team of five to create and set climbing routes for recreational and competition events. Responded effectively to constructive feedback in a fast-paced environment.

## EDUCATION

### M.S. Applied Mathematics, GPA 3.86/4.00

**2023**

University of Colorado Boulder

### B.S. Applied Mathematics, GPA 3.72/4.00 – Cum Laude with Honors

**2022**

University of Colorado Boulder

Minor in Computer Science

- Award: College of Engineering Outstanding Graduate for Research (2022)

## PROJECTS

### Data Analyst @ Bright Block

**April 2025 – June 2025**

- Analyzed bike lane air quality data collected from prototype smart bike lights. Created GIS datasets and data visualizations to support safe cycling advocacy.

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**Undergraduate Research Program Mentor**

**May 2022 – June 2022**

**Boulder Solar Alliance Research Experience for Undergraduates (REU)**

- Developed and led interactive Python workshops for undergraduate research students.